STARCROSS

THE ARTIFACT TEST

A maximum-score route to the successful Earth return

SCORE	MOVES	DEATHS	INSTRUCTIONS
400 / 400	476	0	233,098

Verified story: Z-machine V3, Release 17, Serial 821021; stored checksum matches the calculated checksum. This fresh run reaches the successful ending at 400 points without deaths.

How to use this guide

Play the numbered stages in order. Enter the commands shown in Courier. Repeat `AGAIN` only where a step says to do so or until the stated display or location appears. Before teleporting from the weasel ship, wait 50 turns after dropping the blue disk; this lets the maintenance mouse carry the red disk to the garage.

Route checkpoints

The deterministic replay consumes 288 player inputs and executes 233,098 VM instructions across 57 unique opcodes. Its final score is 400/400 at 476 moves. The route includes optional status checks and a 50-turn wait; the numbered stages compress repeated waits and follow attempts into clear instructions. Story fingerprint: SHA-256 FC3DF0918B56F2C6BC46DF43D9A6E828BCFA24FD8A559AB81789525BF06D3BF8.

Opcode inventory uses 1-based execution sequence numbers and hexadecimal byte addresses. The machine-readable route and complete JSONL trace are stored locally under .agent/regression-routes/ and are intentionally ignored by Git.

01. Plot the course and board the artifact

Take the tape library from the cabin, enter the computer coordinates, and wait for the alien artifact to capture the ship. The route uses a deterministic launch coordinate set.

```
get up
take tape library
east
push red button
read screen
computer, r is 100. theta is 285. phi is 87
computer, confirm
sit
fasten belt
wait (repeat 15 times)
unfasten belt
get up
east
take space suit
take safety line
west
open inner door
out
```

02. Open the alien airlock

Use the sculpture controls to obtain the black rod, then return inside and enter the artifact. The rod opens the outer airlock door.

```
close inner door
wear suit
open outer door
out
examine sculpture
press fourth bump
push tiny column
get black rod
in
close outer door
remove suit
open inner door
up
```

03. Recover the red and yellow rods

Give the tape library to the giant spider for the yellow rod. Take the red rod from the nest on the way in.

```
north
west
north
east
throw library at nest
get library
get red rod
west
south
west
south
open inner door
down
close inner door
open outer door
down
aft
aft
give library
get yellow rod
```

Checkpoint: Both rods are in hand; the spider trades the yellow rod for the library.

04. Repair life support and start the computer

Use the yellow rod in its slot and the red rod in the second red slot. Then install the metal square in the computer access panel and power up the computer.

```
fore
fore
up
close outer door
open inner door
up
up
south
open metal hatch
down
get metal square
put yellow rod in yellow slot
put red rod in second red slot
up
north
down
east
north
east
north
east
south
open panel
put square in slot
close panel
turn computer on
get gold rod
```

Checkpoint: 125 points after collecting the gold rod.

05. Collect the clear, silver, and pink rods

Use the black rod to inspect the projector before dropping it. The silver rod is inside the ray gun barrel. The pink rod is outside the damaged yellow dock; secure the line before crossing.

```
north
east
east
south
west
look in projector through black rod
get clear rod
drop black rod
east
south
south
south
west
get ray gun
look in barrel
get silver rod
east
north
west
open inner door
down
get basket
put all rods in basket
close inner door
wear suit
open outer door
again
down
tie line to suit
tie line to hook
west
get pink rod
east
untie line
up
close outer door
remove suit
drop line
open inner door
```

Checkpoint: 200 points after bringing the pink rod back aboard.

06. Repair the transporter and earn the brown rod

In the lab, use the ray gun and red disk with the globe to produce the blue rod. Give the weasel chief your suit and point to the brown rod. Follow him through the warren, then enter the green airlock and retrieve the violet rod from the weasel ship.

```
up
south
south
east
get red disk
get blue disk
put red disk under globe
drop blue disk
put ray gun on globe
turn dial to 4
turn dial to 1
get all
put all rods in basket
west
south
west
drop red disk
west
north
north
east
east
wait (until the chief appears)
give suit
point at brown rod
follow chief; use AGAIN until Center of the Warren appears
down
close inner door
open outer door
down
west
west
fore
touch skeleton
look
get violet rod
```

Checkpoint: 250 points after the brown rod; 275 after the violet rod.

07. Let the maintenance mouse move the red disk

Drop the blue disk in the weasel ship control room and wait 50 turns before using it. This allows the maintenance mouse to carry the red disk to the garage. Search the trash bin until the green rod appears.

```
drop blue disk
wait (repeat 50 times)
stand on disk
search bin
again
again (repeat until the green rod appears)
get green rod
```

Checkpoint: 300 points after taking the green rod. If you arrive in Green Hall, the red disk has not reached the garage yet; return to the blue disk, wait longer, and try again.

08. Restart the artifact and choose Earth

Carry all rods in the basket to the drive bubble. Insert silver to recover the white rod and restart the stardrive. Shoot the drive bubble three times to reach the control bubble, then insert gold, clear, and the remaining five colored rods.

```
put all rods in basket
north
west
north
north
up
east
south
south
climb tree
up
jump
put silver rod in silver slot
in
get white rod
put white rod in white slot
out
up
jump
shoot drive bubble
again
again
down
put gold rod in gold slot
in
put clear rod in clear slot
put brown rod in brown slot
put green rod in green slot
put blue rod in blue slot
put violet rod in violet slot
put pink rod in pink slot
```

Checkpoint: 350 points after all seven control-bubble rods are inserted.

09. Set the orbital course and finish

Select the solar-system view, cycle the brown spot until Earth is highlighted, and connect the course to the center. Cycle the orbit setting twice to an ellipse, increase speed once, and launch.

```
press large square
press brown spot
again (repeat 3 times, until Earth is highlighted)
press violet spot
again
again
press green spot
again
press blue button
```

Checkpoint: 400 / 400 points, 476 moves, no deaths. The artifact reaches Earth and the successful ending plays.

Executed opcode inventory

One row per unique opcode form and number observed in the completed replay. Counts and first-execution positions come from the final clean trace.

Form	Opcode	Mnemonic	Count	First seq.	First PC
Variable	0x00	call	14,635	1	0x004391
Long	0x14	add	10,709	3	0x0042FF
Long	0x01	je	25,509	5	0x004307
Long	0x15	sub	3,445	6	0x00430B
Short	0x00	jz	39,905	7	0x00430F
Variable	0x01	storew	7,149	9	0x00431A
Short	0x0B	ret	2,456	10	0x00431F
Long	0x0F	loadw	19,493	19	0x004321
Short	0x0C	jump	19,336	22	0x00432F
Long	0x0D	store	16,863	32	0x0043AB
Long	0x0E	insert_obj	394	35	0x0043B4
Long	0x0A	test_attr	6,477	54	0x00440B
Short	0x02	print	732	56	0x006085
Variable	0x09	and	1	58	0x00611A
Variable	0x06	print_num	23	59	0x006120
Long	0x05	inc_chk	7,055	61	0x006132
Long	0x10	loadb	7,077	62	0x006139
Variable	0x05	print_char	6	63	0x00613D
Short	0x0B	new_line	681	87	0x006143
Short	0x00	rtrue	1,743	88	0x006144
Variable	0x08	push	1,968	95	0x005BA0
Variable	0x09	pull	212	97	0x005BA9
Long	0x0B	set_attr	169	100	0x005BD8
Long	0x06	jin	2,202	102	0x005BDE
Short	0x0A	print_obj	187	103	0x005BE2
Short	0x03	get_parent	2,867	106	0x005BEC
Long	0x11	get_prop	2,054	113	0x005C07
Short	0x0D	print_paddr	151	119	0x005C1D
Short	0x02	get_child	2,534	131	0x005C68
Variable	0x01	je	5,878	144	0x005DA1
Short	0x01	get_sibling	2,407	149	0x005DF9
Short	0x05	inc	889	182	0x005E5B
Short	0x01	rfalse	1,574	212	0x007131
Short	0x08	ret_popped	1,624	252	0x005C84
Long	0x04	dec_chk	2,122	306	0x0047D3
Variable	0x04	sread	288	312	0x0047E3
Long	0x07	test	3,350	328	0x004A59
Long	0x03	jg	3,959	333	0x004A5D
Long	0x09	and	1,230	334	0x004A61
Long	0x16	mul	618	344	0x00491C
Variable	0x02	storeb	414	347	0x004928
Long	0x02	jl	2,993	370	0x004EAD
Variable	0x03	jg	30	469	0x005FA1
Variable	0x07	random	491	1,000	0x007BE1
Variable	0x0D	store	133	1,464	0x005500
Long	0x12	get_prop_addr	3,887	1,474	0x0057F5
Short	0x04	get_prop_len	2,410	1,488	0x005ABE
Long	0x17	div	1,383	1,489	0x005AC1
Long	0x0C	clear_attr	117	1,707	0x0068C4
Short	0x03	print_ret	135	1,714	0x0067BA
Short	0x07	print_addr	3	11,254	0x00524E
Long	0x18	mod	13	14,639	0x009428
Short	0x09	remove_obj	212	16,208	0x009ACB
Long	0x13	get_next_prop	875	16,231	0x009D7E
Variable	0x03	put_prop	16	28,214	0x005F26
Short	0x06	dec	13	125,209	0x00D408
Short	0x0A	quit	1	233,098	0x006026